

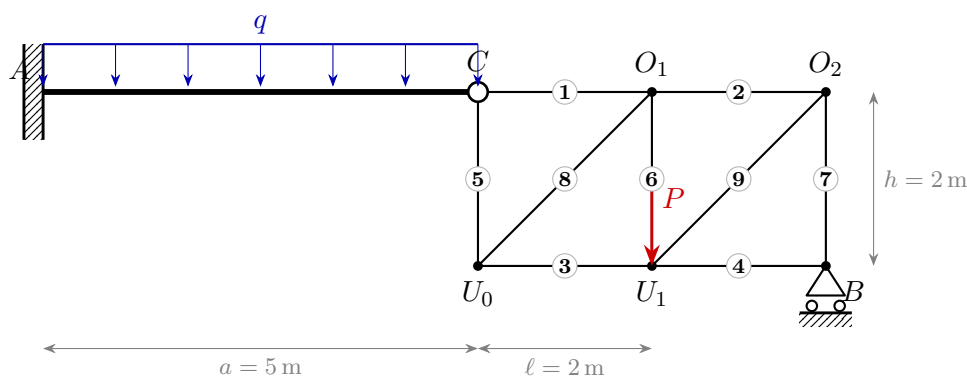
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VU Mechanik – LAWI 100515 (PI)

Exam **Group A**

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Model solution

**1. Static determinacy**Counting criterion $n = r - (3 + v)$ with r support reactions and v hinge conditions:

$$n = r - (3 + v) = 4 - (3 + 1) = 0 \quad \Rightarrow \quad \text{the system is statically determinate.}$$

2. Support reactions and hinge force

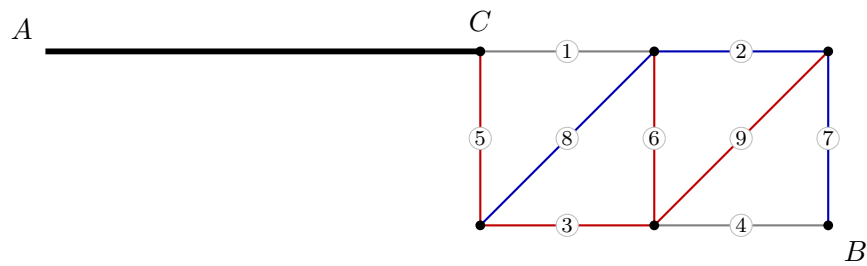
$$A_x = 0 \text{ kN}, \quad A_y = 32.5 \text{ kN}, \quad M_A = 100 \text{ kNm}, \quad B_y = 7.5 \text{ kN}$$

$$C_x = 0 \text{ kN}, \quad C_y = -7.5 \text{ kN}$$

3. Truss member forces

Member	Member force S [kN]	Type
1 (C – O1)	0	zero-force
2 (O1 – O2)	-7.5	compression
3 (U0 – U1)	7.5	tension
4 (U1 – U2)	0	zero-force
5 (C – U0)	7.5	tension
6 (O1 – U1)	7.5	tension
7 (O2 – U2)	-7.5	compression
8 (U0 – O1)	-10.61	compression
9 (U1 – O2)	10.61	tension

red: tension (+) blue: compression (–)



4. Section-force diagrams of the bending beam $A-C$

